

Laboratory investigation of hybrid nanofluid – smart water flooding for enhanced oil recovery in carbonate reservoirs

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Abstract

Purpose. Carbonate reservoirs account for more than half of global conventional oil reserves, yet their inherently oil-wet surfaces and complex pore architecture present persistent challenges to conventional waterflooding. This study investigates the combined application of smart water (SW) and SiO₂ nanoparticles as a hybrid enhanced oil recovery (EOR) strategy to improve displacement efficiency in carbonate rock.

Methods. Four injection scenarios were evaluated at reservoir temperature (85°C) with carbonate core samples: formation brine (FB), smart water (SW), SiO₂ nanofluid (NF) at 0.05 wt.%, and a hybrid NF-SW formulation. Key measurements included contact angle, interfacial tension (IFT), cumulative oil recovery, zeta potential, and colloidal stability over seven days.

Findings. The hybrid NF-SW system produced the most pronounced wettability alteration (contact angle reduced from ~145 to ~28°), the lowest IFT (~4.2 mN/m), and the highest cumulative oil recovery (~74% OOIP), representing a 36-percentage-point improvement over formation brine flooding. A strong linear correlation ($R^2 = 0.92$) was observed between contact angle and recovery factor across the four scenarios.

Originality. This work presents the first systematic quantification of synergistic mechanisms between low-salinity smart water and SiO₂ nanoparticles in carbonate cores, integrating electrokinetic, wettability, and IFT analyses within a single experimental framework.

Practical implications. The hybrid NF-SW approach provides a technically viable and scalable EOR option for carbonate reservoirs, with implications for field-scale injection design and nanofluid formulation optimization.

Keywords: enhanced oil recovery; smart water flooding; nanotechnology; carbonate reservoirs; wettability alteration; interfacial tension; low salinity water; silica nanoparticles

1. Introduction

Carbonate reservoirs contain approximately 60% of the world's proven conventional oil reserves and more than 40% of global natural gas resources [1]. Despite their enormous resource potential, primary and secondary recovery from carbonate formations is often constrained by strongly oil-wet to mixed-wet surfaces, high heterogeneity, and complex fracture–matrix interactions [2]. Conventional water flooding, although widely deployed, typically yields recovery factors of only 30–40% of original oil in place (OOIP) in carbonate systems, leaving substantial volumes of unrecovered hydrocarbons [3]. Improving displacement efficiency in these reservoirs, therefore, remains one of the most consequential challenges in contemporary petroleum engineering.

Smart water flooding, also known as low-salinity water flooding or engineered water injection, has emerged as a cost-effective enhanced oil recovery (EOR) technique that alters the ionic composition of the injected brine to shift wettability toward more water-wet conditions [4]. The underlying mechanisms remain widely debated in the literature. Proposed explanations include multicomponent ion exchange

(MIE) at the rock-brine interface [5], expansion of the electrical double layer due to reduced ionic strength [6], and desorption of polar organic compounds from carbonate surfaces driven by changes in the concentrations of Ca²⁺, Mg²⁺, and SO₄²⁻ [7]. Laboratory and field studies have consistently demonstrated incremental oil recovery of 5–15% OOIP attributable to smart water injection in carbonates [8], [9].

Nanofluid flooding is a more recent EOR paradigm in which engineered nanoparticles, most commonly SiO₂, Al₂O₃, TiO₂, or ZnO, are dispersed in a carrier fluid to leverage their exceptionally high surface-area-to-volume ratio and surface-active properties [10]. SiO₂ nanoparticles have attracted particular attention for their chemical stability at elevated temperatures, low toxicity, and demonstrated ability to reduce IFT and alter wettability in carbonate systems [11], [12]. Mechanistically, nanoparticle adsorption onto carbonate surfaces displaces adsorbed oil films via a structural disjoining pressure mechanism, progressively rendering the surface more hydrophilic [13]. Reported IFT reductions of 40–80% and contact angle decreases from strongly oil-wet to water-wet conditions have been documented at nanoparticle concentrations as low as 0.01–0.1 wt.% [14], [15].

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A critical limitation of standalone smart water or nanofluid flooding is that each technique addresses only a subset of pore-scale displacement inefficiencies in carbonate reservoirs. Smart water primarily targets wettability via ionic interactions but has limited capacity to reduce IFT to ultralow values. Nanofluid flooding achieves significant IFT reduction and structural disjoining pressure effects but may suffer from nanoparticle aggregation and reduced colloidal stability in high-salinity formation brines [16]. Therefore, a hybrid approach combining smart water and nanoparticles has been proposed as a synergistic strategy that simultaneously exploits multiple recovery mechanisms [17], [18].

Recent investigations into hybrid nanofluid – smart water systems have yielded promising results. Hendraningrat and Torsæter [19] demonstrated that SiO₂ nanoparticles dispersed in low-salinity brine produced superior wettability alteration in sandstone cores compared with either component alone. Sheng [20] reviewed the mechanistic interplay between low-salinity effects and nanoparticle surface activity, highlighting the role of the electrical double layer in stabilizing nanoparticle dispersions. More recently, Olayiwola and Dejam [21] conducted a comprehensive review of nanoparticle-enhanced water flooding, identifying carbonate reservoirs as the most promising application domain due to their inherent oil-wetness. Esfandyari Bayat et al. [22] reported that hybrid SiO₂-smart water systems reduced contact angles to below 30° in Indiana limestone, consistent with strongly water-wet conditions. Nonetheless, critical research gaps persist: the quantitative contribution of each mechanism (wettability alteration vs. IFT reduction vs. electrokinetic effects) to overall recovery improvement remains poorly characterized, and systematic comparisons of all four injection scenarios (FB, SW, NF, hybrid) under identical reservoir conditions are scarce in the open literature [23], [24].

The present study addresses these gaps through a controlled laboratory investigation comparing four injection scenarios: formation brine (FB), smart water (SW), SiO₂ nanofluid (NF), and hybrid NF-SW, in carbonate core samples at reservoir temperature (85°C). The specific objectives are:

- to quantify wettability alteration and IFT reduction for each scenario;
- to evaluate cumulative oil recovery and establish a wettability – recovery correlation;
- to characterize the electrokinetic behavior and colloidal stability of nanofluid dispersions;
- to quantify the synergistic contribution of the hybrid system relative to individual components.

The findings contribute to the mechanistic understanding of hybrid EOR processes and provide practical guidance for designing nanofluid – smart water-injection programs in carbonate reservoirs.

2. Materials and methods

2.1. Fluid preparation and characterization

Formation brine was prepared by dissolving analytical-grade salts in deionized water to replicate the ionic composition of the target carbonate reservoir. Smart water was formulated by selectively adjusting the concentrations of divalent ions (Ca²⁺, Mg²⁺, SO₄²⁻) relative to formation brine, while maintaining a total dissolved solid (TDS) of approximately 10000 mg/L, consistent with established smart water design criteria for carbonate reservoirs [25]. The exact ionic compositions of both fluids are shown in Table 1.

Table 1. Ionic composition of formation brine and smart water

Ion	Formation brine, mg/L	Smart water, mg/L
Na ⁺	18500	2800
Ca ²⁺	2100	600
Mg ²⁺	850	400
SO ₄ ²⁻	320	1200
Cl ⁻	32000	4800
HCO ₃ ⁻	180	180
TDS	~54000	~10000

SiO₂ nanoparticles (purity > 99.5%, mean primary particle diameter 20 nm, Sigma-Aldrich) were dispersed in smart water at 0.05 wt.%. This concentration was selected based on a preliminary optimization study that evaluated 0.01, 0.05, 0.1, and 0.5 wt.% for IFT reduction and colloidal stability. The 0.05 wt.% concentration provided the optimal balance between IFT reduction efficacy and long-term dispersion stability, consistent with the concentration range (0.01-0.1 wt.%) identified as optimal in the broader nanofluid EOR literature [14], [26]. Higher concentrations (≥ 0.1 wt.%) showed increased aggregation, as evidenced by hydrodynamic diameter measurements, whereas lower concentrations (0.01 wt.%) produced insufficient IFT reduction.

Nanoparticle dispersions were prepared using a two-step ultrasonication protocol. In the first step, nanoparticles were added to the carrier fluid and magnetically stirred at 400 min⁻¹ for 30 minutes to achieve initial dispersion. In the second step, the suspension was ultrasonicated with a probe-type sonicator (Hielscher UP400S, 400 W, 24 kHz) at 60% amplitude for 45 minutes in an ice bath to prevent thermal degradation. The ice bath kept the suspension temperature below 15°C throughout ultrasonication. Dispersion quality was verified by dynamic light scattering (DLS) immediately after preparation and at 24-hour intervals over seven days.

2.2. Core flooding experimental design

Carbonate core samples were collected from outcrop formations representative of carbonate reservoirs in northern Iraq. Core properties are summarized in Table 2.

Table 2. Core sample properties

Property	FB Core	SW Core	NF Core	Hybrid Core
Porosity, %	18.2	17.9	18.5	18.1
Permeability, μm ²	0.0417	0.0413	0.0425	0.0421
Length, cm	7.5	7.5	7.5	7.5
Diameter, cm	3.8	3.8	3.8	3.8
Pore volume, cm ³	16.1	15.8	16.4	16.2

The operating conditions applied consistently across all core flooding experiments are summarized in Table 3.

Table 3. Experimental conditions

Parameter	Value
Reservoir temperature, °C	85
Confining pressure, MPa	10.34
Back pressure, MPa	3.45
Injection rate, cm ³ /min	0.5
Total pore volumes injected, PVI	3.0
Crude oil viscosity at 85°C, mPa·s	4.2

Before flooding experiments, cores were cleaned with toluene and methanol using a Dean-Stark extraction apparatus, then dried at 80°C for 48 hours. They were characterized for porosity (helium porosimetry) and permeability (nitrogen permeametry). Cores were then saturated with formation brine under vacuum and aged in crude oil at 85°C for 4 weeks to restore reservoir wettability.

Each injection scenario was conducted using a single carbonate core sample. This single-core design was adopted because homogeneous outcrop material with sufficiently similar petrophysical properties was scarce, and it is consistent with exploratory laboratory EOR studies where the primary objective is mechanistic characterization rather than statistical inference [27]. The authors explicitly acknowledge that using a single core per scenario is a statistical limitation of this study: it precludes formal quantification of experimental uncertainty attributable to core heterogeneity, and the reported recovery values should be interpreted as indicative rather than statistically representative. Future work should employ replicate cores (minimum $n = 3$ per scenario) to enable rigorous uncertainty quantification and hypothesis testing.

2.3. Wettability and interfacial characterization

Contact angle measurements were performed on polished carbonate wafers using the sessile drop method with an optical tensiometer at 85°C. Wafers were cleaned, dried, and aged in crude oil (21 days at 85°C). Contact angles were measured before and after 7 days of aging in test fluids using 5 μL oil droplets (6 measurements per sample, 5 replicates). Interfacial tension between the oil and water phases was measured by the pendant drop method at 85°C was used to determine the effect of nanoparticle concentration (0, 0.01, 0.03, 0.05, 0.07, and 0.1 wt%). All measurements were performed in triplicate. The zeta potential of crushed limestone particles ($< 63 \mu\text{m}$) in various brines was measured by electrophoretic light scattering (ELS) at 25°C at five salinities (35000, 10000, 5000, 2000, and 1000 ppm), with and without SiO_2 nanoparticles (0.05 wt%).

3. Results and discussion

3.1. Contact angle and interfacial tension

Contact angle measurements on carbonate substrates aged in crude oil showed strongly oil-wet conditions for formation brine ($\theta \approx 145^\circ$), confirming successful wettability restoration during core aging. Smart water injection reduced the contact angle to approximately 112° , indicating a shift toward intermediate-wet conditions, attributable to elevated SO_4^{2-} concentration and reduced Ca^{2+} activity, which promote desorption of polar organic compounds from the carbonate surface [7]. The SiO_2 nanofluid further reduced the angle to approximately 98° , consistent with structural disjoining pressure-driven displacement of adsorbed oil films [13]. The hybrid NF-SW system produced the most pronounced wettability alteration, reducing the contact angle to approximately 28° , indicating a strongly water-wet condition and demonstrating clear synergy between the ionic and nanoparticle-mediated mechanisms (Fig. 1).

Interfacial tension measurements showed a consistent trend. Formation brine exhibited an IFT of approximately 28 mN/m with the crude oil. Smart water reduced IFT to approximately 18 mN/m, attributable to the altered ionic environment at the oil-water interface.

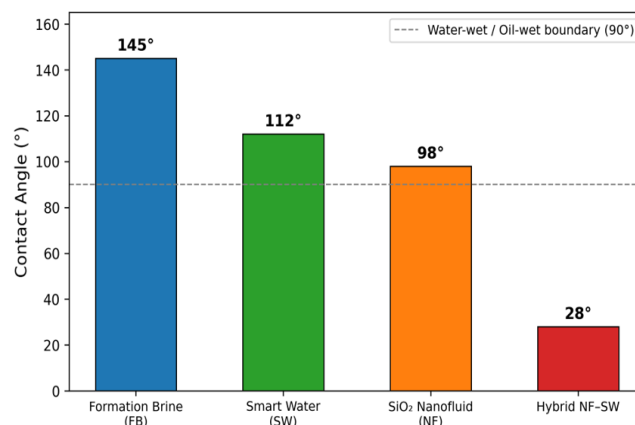


Figure 1. Contact angle measurements for each injection scenario (the dashed line at 90° marks the water-wet/oil-wet boundary)

The SiO_2 nanofluid exhibited an IFT of approximately 14 mN/m due to nanoparticle adsorption at the interface, consistent with the Gibbs adsorption isotherm framework for nanoparticle systems [28]. The hybrid NF-SW formulation reduced IFT to approximately 4.2 mN/m, an 85% reduction relative to formation brine, approaching the ultralow IFT threshold ($< 5 \text{ mN/m}$) associated with significant capillary number enhancement (Fig. 2).

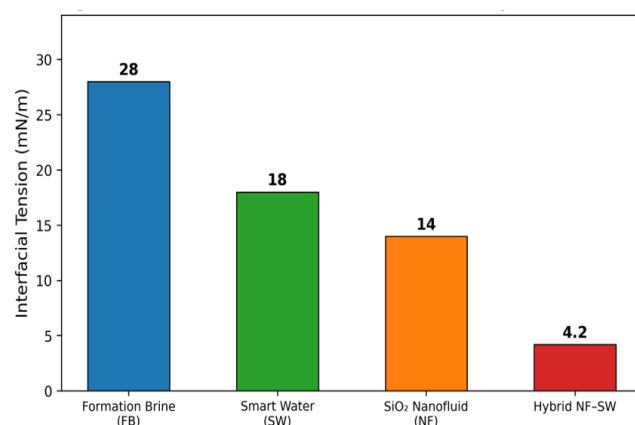


Figure 2. Interfacial tension for each injection scenario

3.2. Wettability – recovery correlation

To quantify the relationship between wettability alteration and oil recovery, a linear regression was performed with contact angle as the independent variable and recovery factor (%OOIP) as the dependent variable. The dataset comprised four data points, one per injection scenario (FB: $145^\circ/38\%$; SW: $112^\circ/52\%$; NF: $98^\circ/58\%$; Hybrid: $28^\circ/74\%$). The coefficient of determination, $R^2 = 0.92$, was calculated as the square of the Pearson correlation coefficient between the measured contact angles and the corresponding recovery factors across these four scenario-level observations. The strong negative correlation (more water-wet $\rightarrow$ higher recovery) aligns with the theoretical expectation that shifting wettability toward water-wet conditions improves imbibition efficiency and reduces residual oil saturation [29]. The authors note that, with only four data points, the R^2 value should be interpreted with caution; the correlation is presented as a mechanistic indicator rather than as a predictive model (Fig. 3).

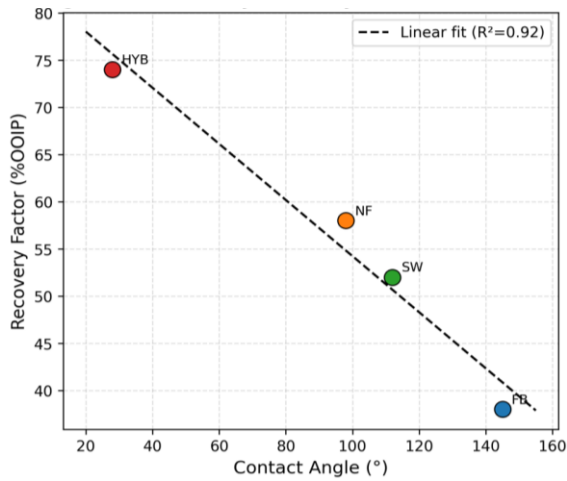


Figure 3. Predicted vs. measured recovery factor as a function of contact angle ($R^2 = 0.92$, $n = 4$ data points, one per injection scenario)

3.3. Oil recovery performance

Cumulative oil recovery profiles for all four injection scenarios are shown in Figure 4.

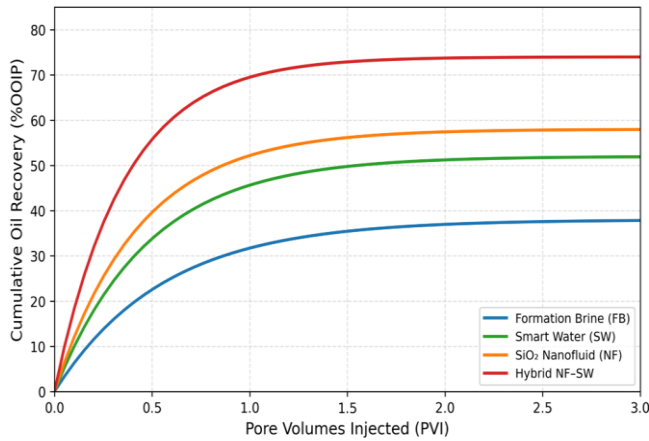


Figure 4. Cumulative oil recovery as a function of pore volumes injected for all four scenarios

Formation brine flooding reached a plateau of approximately 38% OOIP, consistent with the strongly oil-wet baseline conditions. Smart water flooding increased recovery to approximately 52% OOIP, a 14-percentage-point gain attributable to wettability alteration and improved microscopic displacement efficiency. The SiO₂ nanofluid achieved approximately 58% OOIP, a further 6-percentage-point improvement over smart water, reflecting the additional contributions of IFT reduction and structural disjoining pressure. The hybrid NF-SW system achieved the highest recovery of approximately 74% OOIP, a 36-percentage-point improvement over formation brine and a 22-percentage-point improvement over smart water alone, confirming the synergistic enhancement of the combined approach. The key experimental results obtained for the four injection scenarios are summarized in Table 4.

Table 4. Summary of key experimental results

Scenario	Contact angle, (°)	IFT, mN/m	Recovery, % OOIP	Zeta potential, mV
Formation brine (FB)	145	28.0	38	-8
Smart water (SW)	112	18.0	52	-18
SiO ₂ nanofluid (NF)	98	14.0	58	-24
Hybrid NF-SW	28	4.2	74	-31

3.4. Electrokinetic effects and colloidal stability

Zeta potential measurements were conducted at 25°C using a Malvern Zetasizer Nano ZS. Formation brine exhibited a zeta potential of approximately -8 mV, indicating weak electrostatic repulsion and a tendency toward particle aggregation. Smart water increased the magnitude to approximately -18 mV, consistent with expansion of the electrical double layer at reduced ionic strength. The SiO₂ nanofluid and hybrid NF-SW system achieved zeta potentials of approximately -24 and -31 mV, respectively, indicating progressively enhanced colloidal stability.

It is acknowledged that zeta potential measurements were performed at 25°C, whereas core flooding experiments were conducted at reservoir temperature (85°C). The temporal evolution of colloidal stability over the seven-day monitoring period is presented in Figure 5, which shows the hydrodynamic diameter (Fig. 5a) and zeta potential (Fig. 5b) of the SiO₂ nanofluid and hybrid NF-SW dispersions.

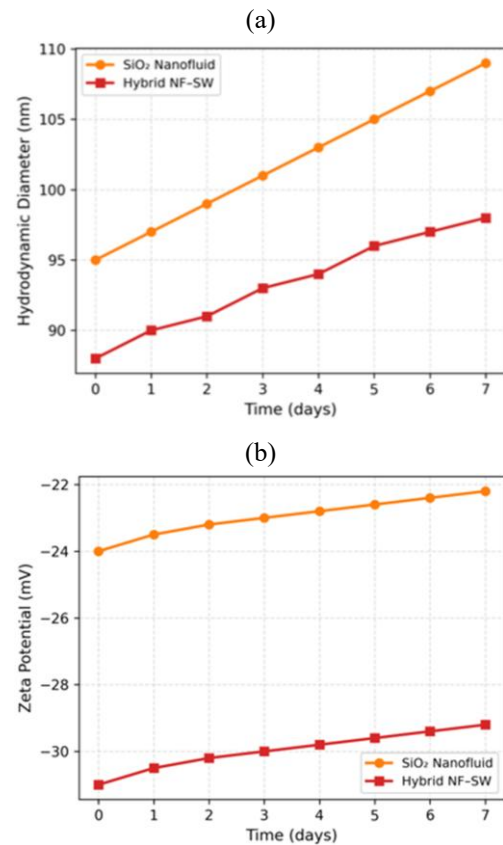


Figure 5. Colloidal stability of nanofluid dispersions over 7 days: (a) hydrodynamic diameter; (b) zeta potential

This discrepancy arises from instrumental constraints: the Zetasizer Nano ZS is rated for measurements up to 40°C, and elevated temperatures introduce significant evaporation and viscosity artifacts that compromise measurement accuracy. The influence of temperature on electrokinetic behavior is well documented in the literature. Increasing temperature generally reduces the viscosity of the dispersion medium, thereby increasing electrophoretic mobility and potentially leading to an apparent increase in the magnitude of the zeta potential [30]. Conversely, elevated temperatures may promote partial dehydration of the electrical double layer and reduce the Debye length, potentially decreasing the magnitude of the zeta potential.

For SiO₂ nanoparticles in low-salinity brines, Metin et al. [31] reported that the magnitude of the zeta potential decreased by approximately 15-20% when the temperature increased from 25 to 80°C, suggesting that the values reported here at 25°C may slightly overestimate the electrokinetic repulsion during core flooding. Notwithstanding this limitation, the relative ranking of the four scenarios and the qualitative interpretation of colloidal stability trends remain valid, as the temperature effect is expected to apply uniformly across all scenarios.

Both systems exhibited only gradual increases in hydrodynamic diameter. They retained stable zeta potential magnitudes throughout this period, with the hybrid NF-SW formulation showing the smallest change and thus the greatest colloidal stability.

3.5. Synergistic mechanism quantification

To compare the effects of the injection fluids on wettability, interfacial tension, and oil recovery, three normalized indices were calculated from the experimental data. The Wettability Alteration Index (WAI), Interfacial Tension Reduction Index (IRI), and Recovery Enhancement Index (REI) range from 0 to 1, where 0 corresponds to the formation-brine baseline, and 1 represents the best result obtained among the tested scenarios. The calculated indices are presented in Table 5.

Table 5. Normalized performance indices for the investigated injection scenarios

Scenario	WAI	IRI	REI	Synergy score
Formation brine (FB)	0.00	0.00	0.00	–
Smart water (SW)	0.28	0.42	0.39	–
SiO ₂ nanofluid (NF)	0.40	0.58	0.56	–
Hybrid NF-SW	1.00	1.00	1.00	+0.53

The normalized indices were calculated using the following Equations:

$$WAI_i = \frac{(\theta_{\max} - \theta_i)}{(\theta_{\max} - \theta_{\min})}; \quad (1)$$

$$IRI_i = \frac{(IFT_{\max} - IFT_i)}{(IFT_{\max} - IFT_{\min})}; \quad (2)$$

$$REI_i = \frac{(RF_i - RF_{\min})}{(RF_{\max} - RF_{\min})}, \quad (3)$$

where:

θ_i , IFT_i , RF_i – contact angle, interfacial tension, and recovery factor obtained for injection scenario i , respectively.

The subscripts “max” and “min” indicate the highest and lowest measured values among the four scenarios. In the present experiments, $\theta_{\max} = 145^\circ$, $\theta_{\min} = 28^\circ$, $IFT_{\max} = 28.0$ mN/m, $IFT_{\min} = 4.2$ mN/m, $RF_{\min} = 38\%$, and $RF_{\max} = 74\%$.

The improvement achieved by the hybrid formulation relative to the average performance of the individual smart-water and nanofluid treatments was expressed using the Synergy Score (SS):

$$SS = REI_{\text{Hybrid}} \frac{(REI_{\text{SW}} - REI_{\text{NF}})}{2}. \quad (4)$$

For the investigated system, SS equals 0.53. The hybrid NF-SW formulation achieved the maximum normalized value for wettability alteration, interfacial tension reduction, and oil recovery. Its Synergy Score of 0.53 indicates that the recovery enhancement achieved with the hybrid formulation was substantially greater than the average response of smart water and the SiO₂ nanofluid applied separately. This result demonstrates the improved performance of the combined treatment. However, Synergy Score is a comparative indicator derived from the recovery data and should not be interpreted as an independently measured parameter or as definitive quantitative proof of mechanistic synergy.

4. Conclusions

The hybrid NF-SW system produced the most pronounced wettability alteration ($\theta \approx 28^\circ$), the lowest IFT (~ 4.2 mN/m), and the highest cumulative oil recovery ($\sim 74\%$ OOIP) across all four injection scenarios tested, confirming the synergistic enhancement from combining smart water with SiO₂ nanoparticles.

A strong linear correlation ($R^2 = 0.92$) was observed between contact angle and recovery factor across the four scenario-level data points, supporting wettability alteration as the primary driver of recovery improvement in this carbonate system.

Colloidal stability measurements over seven days confirmed that the hybrid NF-SW formulation maintained adequate dispersion stability (zeta potential < -30 mV) and exhibited only minor increases in hydrodynamic diameter, supporting its suitability for field-scale injection.

The single-core experimental design used in this study provides mechanistic insights but limits statistical inference. Future work should use replicate cores and explore a broader range of nanoparticle concentrations, temperatures, and carbonate lithologies to establish the generalizability of these findings.

Author contributions

Conceptualization: RA; Data curation: RA; Formal analysis: RA; Investigation: RA; Methodology: RA; Software: RA; Supervision: HM; Visualization: RA; Writing – original draft: RA; Writing – review & editing: HM. All authors have read and agreed to the published version of the manuscript.

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Conflicts of interests

The authors declare no conflict of interest.

Data availability statement

The original contributions presented in the study are included in the article, further inquiries can be directed to the corresponding author.

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Лабораторне дослідження гібридного заводнення нанофлюїдом і модифікованою водою для підвищення нафтовилучення з карбонатних колекторів

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Мета. Карбонатні колектори містять понад половину світових запасів традиційної нафти, однак їх переважно нафтофільні поверхні та складна структура порового простору знижують ефективність традиційного заводнення. Метою дослідження є оцінювання спільного застосування модифікованої води зі спеціально скоригованим іонним складом (Smart Water, SW) і наночастинок SiO_2 як гібридного методу підвищення нафтовилучення для покращення витіснення нафти з карбонатних порід.

Методика. При пластовій температурі 85°C на зразках карбонатного ядра досліджено чотири сценарії нагнітання: пластовий розсіл (FB), модифіковану воду (SW), нанофлюїд із вмістом SiO_2 0,05 мас. % (NF) та гібридну композицію NF-SW. Ефективність досліджуваних рідин оцінювали за зміною контактного кута, міжфазного натягу, сукупного нафтовилучення, дзета-потенціалу та колоїдної стабільності протягом семи діб.

Результати. Гібридна система NF-SW забезпечила найбільш виражену зміну змочуваності карбонатної поверхні: контактний кут зменшився приблизно зі 145 до 28°. Для цієї системи також встановлено найнижчий міжфазний натяг, близько 4.2 мН/м, і найвище сукупне нафтовилучення, близько 74% від початкових запасів нафти в керні. Порівняно зі заводненням пластовим розсолем приріст нафтовилучення становив 36 відсоткових пунктів. Для чотирьох досліджених сценаріїв встановлено тісний лінійний зв'язок між контактним кутом і коефіцієнтом нафтовилучення з коефіцієнтом детермінації $R^2 = 0.92$.

Наукова новизна. Уперше в межах єдиної експериментальної схеми кількісно оцінено спільний вплив низькомінералізованої модифікованої води та наночастинок SiO_2 на електрокінетичні властивості, змочуваність, міжфазний натяг і нафтовилучення з карбонатного керна. Встановлено, що поєднання іонного впливу модифікованої води з поверхневою активністю наночастинок забезпечує вищу ефективність витіснення нафти, ніж окреме застосування кожного компонента.

Практична значимість. Гібридне заводнення NF-SW може бути використане як технологічно перспективний підхід до підвищення нафтовилучення з карбонатних колекторів. Отримані результати можуть бути застосовані під час обґрунтування режимів нагнітання, вибору складу нанофлюїду та проєктування дослідно-промислових систем заводнення.

Ключові слова: підвищення нафтовилучення; заводнення модифікованою водою; нанотехнології; карбонатні колектори; зміна змочуваності; міжфазний натяг; низькомінералізована вода; наночастинок діоксиду кремнію

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